

Response to reviewers

Manuscript: *Fairness Auditing and Bias Mitigation in ML: A Technical Workflow*

The authors thank the editor and reviewers for their comments. The revised manuscript has been reframed as a technical workflow for fairness auditing and bias mitigation, rather than as a holistic solution to AI ethics. We also added explicit limitations on data provenance, historical labels, Region as a proxy attribute, synthetic age, fairness–performance trade-offs, aggregate statistical testing, and a sample Model Card.

Editor

Comment 0.1 — Results, claims, and limitations should be revised so that the manuscript does not overstate the ethical contribution.

Reply: We revised the title, abstract, methodology, results, and conclusion to frame the contribution as an applied technical workflow for auditing and mitigation. The revised text now states that the workflow does not solve the broader ethical and legal questions raised by predictive systems in criminal justice, and that the models are not suitable for operational legal decision-making without further validation.

Reviewer 1

Comment 1.1 — The synthetic age variable requires clearer explanation and sensitivity analysis.

Reply: We clarified that synthetic age is not observed defendant age and should not be interpreted as demographic truth. The manuscript now explains the normal and uniform generation strategies and states that the uniform-distribution results are included in the appendices as a sensitivity check.

Comment 1.2 — The manuscript should include statistical tests for the performance claims.

Reply: We added a statistical comparison table. Because the re-executed notebooks produced confusion matrices that did not exactly match the manuscript matrices, we followed a conservative approach and used aggregate two-proportion tests based on the manuscript confusion matrices. We explicitly state that these are not paired McNemar tests and that bootstrap confidence intervals over paired predictions were not used for the manuscript claims.

Comment 1.3 — The fairness–performance trade-off should be stated more directly.

Reply: We revised both experiment discussions and the conclusion to describe the trade-off directly. The drug trafficking model reduces FNR but substantially increases FPR and reduces accuracy. The petty theft model improves recall and F1 but reduces accuracy and precision.

Comment 1.4 — The manuscript should include a Model Card.

Reply: We added a sample Model Card in the appendix. It documents intended use, out-of-scope use, data, labels, proxy protected attributes, fairness metrics, known limitations, and monitoring recommendations.

Comment 1.5 — Language editing is needed.

Reply: We corrected several typos and grammar issues, including misspellings, quote mismatches, duplicated phrasing, and overclaiming in the results and conclusion.

Reviewer 2

Comment 2.1 — Typos and incomplete references should be corrected.

Reply: We corrected visible typos such as “Fairness”, “confusion”, “Foreigner”, “Effective Hearings”, “petty theft”, and “drug trafficking” in the revised manuscript. We also corrected inconsistent quotation marks and several grammatical issues.

Comment 2.2 — Fairness should be defined explicitly.

Reply: We revised the key definitions section. Fairness is now defined operationally as an empirical property of model errors across groups, with emphasis on FNR, FOR, and TPR. We also added an example of regional FNR disparity.

Comment 2.3 — The use of Region as a protected attribute should be justified and limited.

Reply: We now describe Region as a geographic proxy attribute. The revised manuscript explains that Region is useful for auditing territorial disparities but is not a substitute for race, ethnicity, socioeconomic status, gender, real age, or other individual-level protected attributes.

Comment 2.4 — The contribution should be positioned beyond merely using fairness enhancers.

Reply: We repositioned the contribution as a workflow that combines data review, baseline auditing, mitigation, interpretability, disparity analysis, statistical comparison, and documentation. The revised manuscript emphasizes the audit process and its limitations, not only the use of AIF360, Fairlearn, SHAP, or Aequitas.

Reviewer 3

Comment 3.1 — The manuscript should discuss data provenance and label limitations.

Reply: We added text explaining that the data come from historical DPP records and that the favourable/unfavourable labels are practical simplifications of legal outcomes. The revised manuscript states that these labels do not establish legal correctness, defence quality, or absence of historical bias.

Comment 3.2 — The manuscript should acknowledge possible laundering of judicial bias.

Reply: We added this limitation in the data and conclusion sections. The revised manuscript now states that models trained on historical judicial records can reproduce and potentially launder historical biases present in case outcomes.

Comment 3.3 — The generalisation claims should be limited.

Reply: We revised the conclusion to state that the workflow is a practical tool for identifying measurable error disparities, testing mitigation strategies, documenting limitations, and informing whether a model should be further developed, restricted, or rejected. We also added external and prospective validation as future work.

Comment 3.4 — The framing should be changed from a holistic ethics framework to an applied technical audit.

Reply: We changed the title and revised the abstract, methodology, and conclusion accordingly. The manuscript now consistently describes the contribution as a technical fairness-auditing and bias-mitigation workflow.